Individual Demand for Building State Effectiveness

Yetsedaw Bekele, Harry Dienes, Daniel Rogger and Giulio Schinaia*

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Abstract

Training and professional development are essential for building an effective workforce. Yet where benefits of capacity building accrue to society, such as with public servants, individuals may under-invest in professional development relative to the social optimum. This paper presents evidence on the extent of demand for professional development in public administration. We elicit incentive-compatible willingness-to-pay for professional development opportunities among civil servants in Ethiopia. Our findings reveal prevalent low demand for professional development activities key to building state capacity. The median respondent values access to a professional development activity as much as a packet of pens. However, demand is highly elastic, and almost all public officials are willing to undertake training for a small subsidy. An ongoing follow-up experiment aims to disentangle whether the low demand is driven by low private returns to training or by externalities that respondents fail to internalize.

Keywords: bureaucrats, public administration, training

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1 Introduction

State capacity is a central determinant of economic development [Besley and Persson, 2009; Acemoglu, García-Jimeno and Robinson, 2015; Fergusson, Molina and Robinson, 2022]. Despite its established importance, our understanding of the professional development of public servants, a key feature of strengthening state capacity, remains elusive [Grindle and Hilderbrand, 1995; Azulai et al., 2020].

Public servants are both a technical asset in themselves, transforming public policy into government action, as well as managers of other personnel and public resources. Given their extended tenures, professional development of existing public servants is a key avenue for strengthening the state and its ability to deliver. However, the social benefits of an individual public servant's capabilities may not enter into their investment decisions regarding capacity development. Like actors in private organizations, they may under-invest in their capabilities relative to the level that optimizes organization-level objectives [Konings and Vanormelingen, 2015; Adhvaryu, Kala and Nyshadham, 2023; De Grip and Sauermann, 2012]. Unlike private actors, these under-investments have direct implications for state capacity. Together, these constraints imply that leaving the decisions for investing in state capacity to individual public servants may lead to lower state capacity than the social optimum.

This paper investigates how public servant's perceive capacity building in the public sector and measures their demand for professional development activities. To build a picture of these demands that is both precisely identified but also broadly relevant, we combine evidence from a field experiment within a particular public service context with novel survey data from an international initiative to survey public servants.

To better understand the demand for public service capacity building, we designed a framed field experiment akin to a market mechanism to measure it among a sample of public sector workers in Ethiopia. This framed experiment was implemented with public servants from the Civil Service University in Addis Ababa, who are are pursuing further education in the capital and come from subnational government offices. We develop an incentive-compatible experimental measure of demand for individual capacity building among public servants by implementing a Becker-Degroot-Marshak (BDM) mechanism in

the public administration. We offered the opportunity to pay (or be paid) to attend executive training sessions, professional coaching, and work shadowing. We design these professional development opportunities with input from focus groups, and they were offered by the country's leading management institute in partnership with reputable international organizations.

We find a low willingness to pay for such capacity-building opportunities. The median demand for a day's training is equivalent to that for a packet of pens. Yet demand is also highly elastic, implying a degree of responsiveness to centralized efforts to subsidize capacity building. We find a distinctive kink in the demand curve at 0, implying lower inelasticity of demand for a subset of public officials.

To understand the drivers of demand in this context, we conduct topic modelling on the free response answers given by our respondents explaining why they bid as they did. We find suggestive evidence that the more focused on individual payoffs an official is, rather than on the wider social benefits of his capacity building, the lower their willingness-to-pay.

We anchor these results in survey evidence public sector administrations across 26 countries. Using new data from the Global Survey of Public Servants [Schuster et al., 2023], we describe public servant's own perspectives across varying contexts, drawing on over 1 million individual respondents from over 1,100 different institutions. This comparative effort indicates that our results from Ethiopia are consistent with similar concerns from a broad range of settings.

A large literature on workplace training in private sector settings [Blundell et al., 1999] finds that training and other professional development interventions can improve sales, profits, and management [Cusolito, Darova and McKenzie, 2023; Iacovone, Maloney and McKenzie, 2022; Martins, 2021] and improves labour market outcomes [Alfonsi et al., 2020]. Empirical work supports the view that workplace training increases productivity more than wages [Konings and Vanormelingen, 2015; Dearden, Reed and Van Reenen, 2006], including recent experimental research [De Grip and Sauermann, 2012; Adhvaryu, Kala and Nyshadham, 2023]. The tension between the relative benefits of training for employers and workers forms a key part of this literature.[1] Theoretically, Acemoglu and Pischke [1998, 1999] argue that firms

[1]Recent experimental evidence from Burundi shows that employers under-invest in

are able to benefit from investing in training to the extent that there are labour market imperfections that inhibit turnover and allow a compressed wage structure. Firms will invest in worker training to the extent that labour markets are not fully competitive, allowing for consequent productivity gains to be captured by the firm. Conversely, workers can expect to benefit from investing in their professional development in labour markets with fewer frictions.

A nascent literature identifies positive productivity and performance impacts of training in public administration with experimental evidence [Azulai et al., 2020; Jakobsen, Jacobsen and Serritzlew, 2019; Jacobsen et al., 2022; Kroll and Moynihan, 2015] and panel data [Seidle, Fernandez and Perry, 2016], suggesting that training can have positive impacts in the public sector. However, job move transaction costs are particularly high for public sector jobs, characterized by skills that may be less transferable to similar private sector positions. This implies the incentive for workers to invest in capacity building will be particularly low.[2]

There is very limited evidence on the drivers of individual demand for capacity building. This is despite the fact that the success of workplace development itself may depend on co-investment of time and effort by those undergoing development programs.[3] We are not aware of studies that mea-

training of their employees because of labour market frictions and spillovers to other firms that reducing employer's incentives [Cefala et al., 2024].

[2]Such analysis is complicated by the fact that while compression in public sector wage structures [Gindling et al., 2020] suggests that public employers will be able to capture more of the the gains from training than employees, the lack of profit signals changes the objective function of bureaucracies and their managers. Existing evidence is not clear on the the size of the wage return to training in the public sector, though an analysis of household panel data to compare the training in the public and private sectors in the USA and UK suggests large differences between the two countries [Méndez and Sepúlveda, 2016]. The USA, ranked highest for bureaucratic quality, has a large return to employer-provided training in the public sector of 6 percent, against a private sector return of 2 percent. In the UK, however, the return in both sectors are smaller, and the order reverses: a 0.9 percent increase in the private sector and a 0.7 percent increase in the public sector.

[3]Theoretically, training demand may be impacted by the degree employees expect their actions to lead to a corresponding wage growth, the sense that an individual's capabilities are influential, and whether official's believe training contributes to the achievement of professional goals. Using German household panel data, Caliendo et al. [2022] show that a higher internal locus of control leads to greater take up of training, owing to increased expectations of wage rises rather than actual wage rises. Meta-analyses of observational

sure the demand for professional development or capacity building inside public administration, despite the theoretical reasons to believe individual demand will be important to the development of state capacity. As such, we contribute to the scientific understanding of public sector capacity by measuring willingness to pay in a controlled setting and embedding our findings in a comparative cross-national analysis regarding what may drive such demand structures.

The paper continues as follows. Section 2 presents the results of a demand estimation exercise undertaken amongst public servants from the Government of Ethiopia. Section 3 explores the determinants of willingness-to-pay from survey data collected in parallel with the demand estimation exercise. Section 4 provides comparative evidence on the demand for training from Ethiopia and across the world. Section 5 concludes.

2 Estimating Demand for Capacity Building

2.1 Capacity Building in Ethiopia's Public Service

Ethiopia is Africa's second most populous country, home to 100 million people or 10% of the population of Africa. It has a large public service, employing roughly half of paid employees in the country [Baig et al., 2021]. Corruption is less prevalent than in most of Africa, but government effectiveness is seen as relatively weak overall [Kaufmann and Kraay, 2023].

Like many public administrations, capacity building initiatives are a pillar of Ethiopia's public service. A 2024 survey of Ethiopian public servants indicates that 43 percent had received training in the last 12 months, among whom 97 percent generally receive training at least once a year. Among respondents who attended training, 72 percent received internal training, while 23 percent went on external courses.[4]

The survey indicates that the training public servants receive appears to be

studies in psychology and organizational behaviour have found workplace learning is more successful when tied to clear professional goals [Kraiger and Ford, 2021] and that motivation and work climate are an important predictor of knowledge transfer from training to the workplace [Ford, Baldwin and Prasad, 2018].

[4]Further details on the surveys of public servants discussed in this section are provided in the Appendix Annex A.

broadly public-sector specific. Among public servants who had received training in the previous 12 months, the most prevalent topics of study were 'relevant laws and regulations' (73 percent), 'administrative processes' (32 percent), 'ethics' (29 percent) and 'core ministry subject matters' (27 percent). While public sector organizations understandably provide their employees with *public-sector specific* knowledge rather than general competencies, this reduces the transferability of the training to other settings and amplifies the benefits to the wider public service.

Respondents considered the training they did receive as relevant to their work (74 percent saying 'very relevant') and improved their workplace productivity (92 percent). These results are highly consistent across sectors, such as health, education, and finance, and across tier of government, whether federal, regional, or local.

Yet the survey also pointed towards the fact that extrinsic incentives for undertaking training were also limited within the public service, despite its perceived effectiveness on productivity by public servants themselves. Only 56 percent of respondents were confident that they would be promoted if they performed well in their public service roles, reducing the incentive to invest in public-service specific capabilities. Meanwhile, continuously developing job specific skills continues to be subordinate to formal education levels, with educational background seen as very important, scoring 4.8 out of 5 on an index of what predicts promotions, whereas job-specific skills averaged 3.0, equivalent to being 'neutral' about the impact on promotions.

2.2 An Incentive-compatible Mechanism to Measure Demand

To measure individual public servant demand for professional development activities within this setting, we created a market for capacity building activities in Ethiopia's public service through a framed field experiment. We focus on three of the most frequently offered activities: 1) executive education; 2) professional coaching; and 3) work shadowing. To ensure the programs offered are of some value, we build on feedback from focus groups to offer products with valued characteristics, run by the pre-eminent management institute in the country, in partnership with international organizations, and

with low costs of participation.[5]

We work with public servants exploring professional development opportunities at the Civil Service University in Addis Ababa, Ethiopia. Their characteristics are similar to the broader public service, as described in Table 1, where we compare their demographics to those of a representative survey of Ethiopian public servants undertaken in 2016. As in the wider public service, participants in our study are dominantly men, in their mid-thirties, who have worked in the public service for almost a decade. They typically have an undergraduate degree already, and are pursuing a further degree at the Civil Service University. Similar to the distribution of the service, they are dominantly from subnational entities, with roughly half from the lowest tier of the Federal Government—the *woreda*.[6]

To elicit willingness to pay for professional development, we design a Becker-DeGroot-Marshak (BDM) mechanism adapted from the recent literature to our setting [Berry, Fischer and Guiteras, 2020; Cole et al., 2020; Burchardi et al., 2021]. BDM mechanisms have been used to measure willingness to pay for a wide variety of products [Berkouwer and Dean, 2022] and have begun to be used to measure demand for services such as education and training [Maffioli, McKenzie and Ubfal, 2022; Berry and Mukherjee, 2019; Abebe et al., 2024].

Our mechanism asks participants the highest amount they are willing to pay for each of three development opportunities (coaching, shadowing, training), the order of which is randomized. The bids for all three opportunities are confirmed with the subjects, with all three bids displayed simultaneously. These first bids are then tested with a series of questions that probe whether they would in fact still like to purchase the course for a small increment in price. They are also reminded that they will in fact pay for the course if the randomised price they receive turns out to be below their bid.[7] Both of the

[5]Further details on the marketing of the opportunities, and the wider protocol used for the experiment, is provided in the Appendix Annex B.

[6]While we cannot observe a difference in objective characteristics from our sample relative to the representative survey of the civil servants, we cannot rule out that selection at Civil Service University may be driven by other factors that we did not measure. For example, if those at Civil Service University were more motivated than the average civil servants and this motivation led them to enrol for postgraduate studies, we may expect that they may also be willing to pay more for continuous professional development.

[7]While all subjects consented to participate, 13 out of 512 did not have a method of

Table 1: Descriptive Statistics

	Benchmark (2016) (N=2,164)	Sample (2023) (N=498)
Female	0.178 (0.383)	0.181 (0.385)
Age	35.178 (8.553)	32.368 (5.472)
Wage (Birr per month)		
0-2,000	.	10 (2.2%)
2,001-5000	.	83 (17.9%)
5,001-10,000	.	308 (66.5%)
10,001-20,000	.	58 (12.5%)
20,001-50,000	.	4 (0.9%)
Work Experience in Civil Service (years)	13.063 (8.549)	9.055 (5.773)
Education		
Grade 8 completion	2 (0.1%)	0 (0.0%)
Grade 10 completion	4 (0.2%)	0 (0.0%)
Grade 12 completion	2 (0.1%)	5 (1.0%)
Diploma / TVET / Post-high school	376 (17.7%)	4 (0.8%)
Undergraduate degree	1,533 (72.2%)	443 (89.0%)
Masters degree	201 (9.5%)	46 (9.2%)
PhD	4 (0.2%)	0 (0.0%)
Program Engaged		
Bachelor Degree	.	33 (6.7%)
Masters	.	455 (92.1%)
Ph.D.	.	6 (1.2%)
Administrative Tier		
Federal	285 (13.2%)	67 (13.6%)
Regional	443 (20.5%)	132 (26.9%)
Zone	0 (0.0%)	60 (12.2%)
Woreda/Sub city	1,436 (66.4%)	232 (47.3%)
Office in Addis Ababa	.	0.245 (0.430)

Note: The 2016 Benchmark is based on the Ethiopian Civil Service Survey (CSS), a representative sample of the Ethiopian Civil Service. The 2023 Experimental Sample is drawn from respondents of the artefactual field experiment attending Civil Service University (CSU) winter session classes. Standard deviation or percentage of category (%) shown in parentheses.

payment available and so did not engage in the framed field experiment, leaving a sample

probing questions are to ensure fidelity and allow participants to carefully consider their valuation.[8]

The participant is then offered a randomly selected opportunity and price.[9] If the offer is lower than the stated maximum willingness to pay for the specific course, the participant purchases the opportunity at the randomized price.[10] If the price drawn is higher, the participant cannot purchase the opportunity. This setup enables an incentive-compatible elicitation of an individualized willingness to pay for each opportunity for every participant. Since the participant does in fact buy the item for real money, but only if the randomised price is less than or equal to the bid, participants are incentivised to reveal their true valuation.

The distribution of prices included negative prices, such that participants would be paid to attend the training. If subjects state a valuation of 0 Birr in their initial valuation, the respondent was then asked how much would be the minimum amount they would need to get paid in order to take part in the activity. We incorporate negative prices—akin to direct financial subsidies—in keeping with a common practice where public servants obtain *per diems* when attending some training, and enables us to measure demand in a context of widespread subsidies.[11] The extension to negative pricing is an important feature of our study, where public servants may not value the time spent in training as highly as their opportunity cost of time, irrespective of the social benefits of the consequent capabilities they may obtain.[12]

of $n = 499$. One of these then stopped after the trial exchange for the pen.

[8] We first ran a practice with participants to familiarise them with the mechanism by offering the chance to buy a pen using the BDM process. The willingness to pay for the pen forms a benchmark to assess the results for the development opportunities.

[9] While the price for each participant was pre-randomised, it was not revealed to either the participant or the enumerator until after the bids for all courses had been made.

[10] All bar four participants paid the relevant price when they were required to do so.

[11] A negative price is equivalent to Willingness to Accept (WTA) in the absence of endowment effects.

[12] The experiment was executed as planned, achieving high implementation fidelity. The process adhered closely to our original design, and we were able to gather the intended data effectively.

2.3 The Demand for Capacity Building

Figure 1 shows the inverse demand curve for all subjects, showing the share of subjects that have a willingness to pay less than p, where $p \in [-300, 700]$. The median offers for coaching, shadowing and training were 80, 90, and 100 Birr, respectively.[13] During the trial item elicitation, the median offer for a pen was 15 Birr. Equivalently, the value of the professional development opportunities was worth around 5 to 6 pens to our study participants. Relative to the median income we assess for the participants, these offers are approximately 1% of median monthly income. Only 2% of participants would pay more than 10% of monthly income for any of the development opportunities we offer.

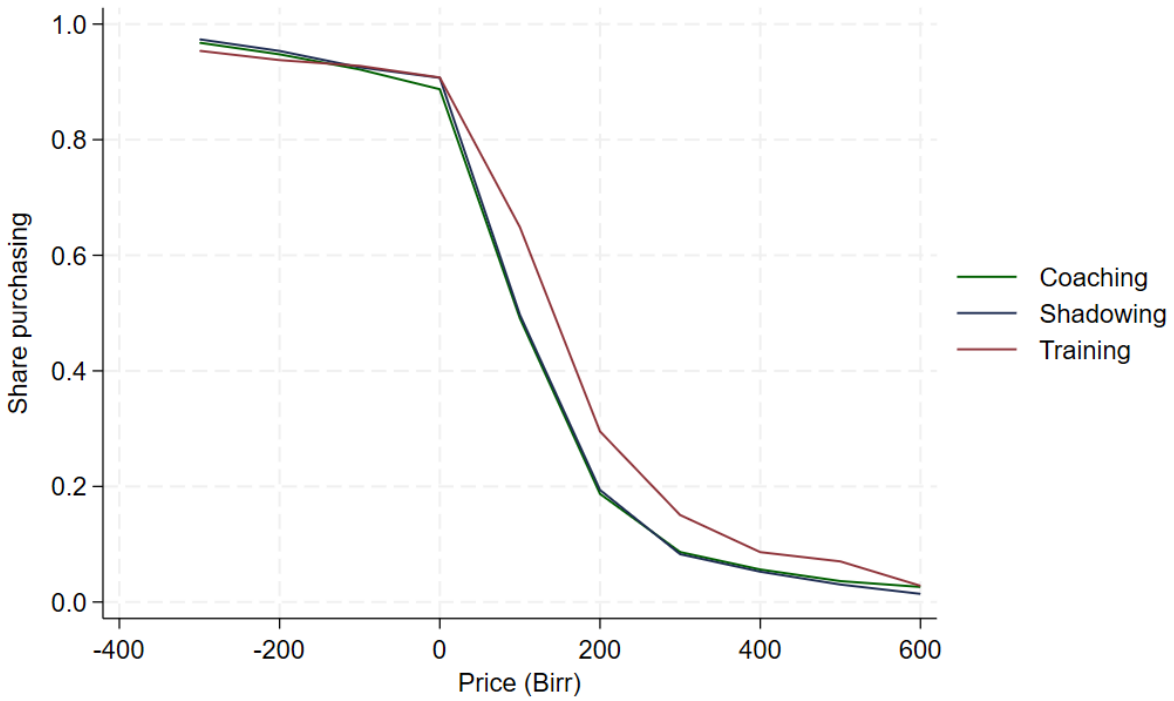


Figure 1: Inverse Demand Curve for Coaching, Shadowing and Training

[13] At the time of the study, the nominal exchange rate was 60 Ethiopian Birr (ETB) = 1 US Dollar, such that these were equivalent to USD1.40, USD1.60 and USD1.80 respectively.

The inverse demand curve for training indicates that there is slightly more demand for this form of professional development than for coaching and shadowing, the inverse demand curves of which closely mirror each other. However, despite the difference in the distributions between training and both coaching and shadowing being significant in Kolmogorov-Smirnov tests ($p = 0.00$), the substantive difference in demand between opportunities is equivalent to the value of a pen for our participants. As such, our findings seem relatively equivalent across types of capacity building.

Though the demand for training is generally low, it is however very elastic—with all three inverse demand curves indicating a high degree of responsiveness to small variations in price. Small deviations in the cost—which is financial here, but is a proxy for wider effort and other costs—imply a large shift in the uptake in capacity building activities. Akin to the literature on socially desirable goods (such as insecticide treated nets) facing low demand at even small positive prices, public servants are highly price-sensitive to investing in their capabilities [Cohen and Dupas, 2010; Mobarak et al., 2012]. Such sentiments are reflected in the Ethiopian Survey of Public Servants 2024, where 82 percent of respondents expressed that there was insufficient *subsidized* training provided to them.

Figure 1 suggests there exists a long tail of positive demand, with roughly 10% of participants willing to pay over 450 Birr for development opportunities. We can also see that the elasticity of demand falls amongst this top group, indicating that there is a small subset of public officials who are highly motivated by gaining capacity building opportunities.[14] This top 10% of respondents have on average more years of experience (10.7 years compared to 8.9 years, $p = 0.026$), higher monthly salaries (an average of 9,000 Birr compared to 7,700 Birr, $p = 0.052$), and are older (34.0 years compared to 32.2 years, $p = 0.030$)—though gender and government tier do not show significant differences.

At the bottom end of the demand curve, there exists a small group of public (6%) that would not pay anything to undertake further professional development. A distinctive kink in the demand curve at a zero price reflects the salience of payment versus subsidy of capacity building in the public service environment. The elasticity of demand falls significantly at a price of zero,

[14] However, none of the participants bid a price high-enough to cover their per capita training cost. As such, in a market setting no training would occur.

indicating that small subsidies are almost as effective as large ones in inducing take up of capacity building. Conversely, such a drop in elasticity also implies there is a small set of public officials for which it is 'expensive' to induce them to take up capacity building activities, since the last few percent of officials in our data require large subsidies to initiate take up. These respondents do not exhibit particular descriptive differences to the rest of the sample, with no difference in gender, age, years experience, government tier or even wage.

Seeing financial investment as a proxy for broader investments in professional development, the overall picture from our experiment is that marginal demand for professional development amongst public servants in Ethiopia is low. Though there is a small group of officials with a higher, and less elastic, demand for capacity building, even these officials make offers that are insufficient to cover the per capita cost of the capacity building initiatives.

3 What Influences the Demand for Capacity Building?

Individual public servants are more likely to demand capacity building if they believe it will strengthen their future prospects, or if they are motivated by its impacts on the broader public good. Conversely, if they do not perceive effective returns to capacity building in their careers, or do not weigh the social benefit of these activities, they will be less willing to invest in them.

To understand what drives the choices around professional development of the public servants in our study, we asked each participant to write a short reflection on what drove their pricing decisions, and the value they see in continuous professional development for their career.[15] For example, quoting from one response, "it is very important to improve my existing skill and experience for future work efficiency in my life". Another respondent stated that, "Ethiopia needs skillful leaders in the government offices. If I get this chance I hope I will contribute to the development of the society that need my professional knowledge."

On average, respondents agreed that the training provided useful opportuni-

[15]Given the need to drop some responses given the quality of the narrative provided, this section uses a subset of 87% of respondents used in the analysis in section 2.

ties for skill-building. The reflections rarely distinguished between types of capacity building. Rather, the passages are expressions of the respondent's view of capacity building overall. To provide a further sense of the tone of the reflections, Figure 2 presents a word cloud from the responses.

skill
performance
develop
service
capacity
skills
future
organization
experience
helps
public
increase
career
job
life
management
enhance
building
time
continuous
society
community
opportunity
opportunities
institutions
developed
employees
office
practice
motivation
activities
attitude
country
change
leadership
information
ethiopia
individual
customer
improves
research
trainings
continues
team
institution
day
civil
win
build
achieve
servants
social
personal
sharing
satisfaction
chance
experiences
evaluation
manage
serve
additional
servant
trading
quality
understanding
effective
developing
communication
coaching
profession
workers
system
people
provide
understand
improvement
delivery
government
students
practical
ability
awareness
gaps
administration
participation
organizations
university
technology
process
support
migration
achievement
profesionalcompetent
individuals
sector
short
train
effectively
plan
income
price
goals
world
gain
game
encourage
services
solve
programs
confidence
gap
create
goal
kinds
objective
addition
employee
share
cost
enables
confidential
enable
field
attend
idea
sectors
success
learn
planning
deliver
provided
trainee
productive
behavior
performing
experts
leader
mission
motivate
motivated
additional
satisfied
expected
expert
analysis
continue
helped
trained
values
types
add
grade
organizational
adoptions
improving
private
participate
perform
leaders
tasks
governance
sustainable
adoption
forever
helpful
facilitate
acceptable
typewriter
promote
successful
increasing
serving
attitudes
developed
developing
managerial
monitoring
economic
promoting
square
carrier
professions
happy
administrative
country's
people's
development
worker
community
benefits
academic
education
activity
competitive
procedures
identify
employees
position
feature
related
office
practice
giving
culture
meet
duties
wants
understood
pertains
essential
federation
external
income
plan
past
finally
price
goals
world
benefit
thinkers
employed
human
accepted
encourage
confidence
programs
solve
services
gap
enhanced
lead
goal
kinds
objective
addition
share
cost
play
duo
employee
manager
strengthen
environment
efficiently
customer
providing
interpersonal
relationship
aware
body
agriculture
improves
learning
research
hope
level
trainings
update
continues
means
bulls
manner
project
experiences
equal
market
chance
evaluation
manager
provided
trainee
productive
behavior
performing
experts
capability
leader
mission
motivate
motivated
additional
satisfied
expected
expert
analysis
continue
helped
trained
values
types
add
grade
developed
developing
managerial
monitoring
economic
promoting
square
carrier
professions
happy
administrative
organizational
adoptions
improving
private
participate
perform
leaders
tasks
governance
sustainable
adoption
forever
helpful
facilitate
acceptable
typewriter
promote
successful
increasing
serving
attitudes
quality
effective
understanding
organization
enhance
satisfaction
helps
personal
sharing
future
skills
public
capacity
ethiopia
individual
customer
improves
research
trainings
continues
team
institution
day
civil
win
build
achieve
servants
social
leadership
change
country
management
life
attitude
activities
motivation
job
increase
career
continuous
time
practice
office
employees
developed
institutions
society
community
opportunity
opportunities
service
performance
develop
experience
coaching
communication
profession
workers
system
people
provide
understand
improvement
delivery
government
students
practical
ability
awareness
gaps
administration
participation
organizations
university
technology
process
support
migration
achievement
profesionalcompetent
individuals
sector
short
train
effectively
plan
income
price
goals
world
gain
game
encourage
services
solve
programs
confidence
gap
create
goal
kinds
objective
addition
employee
share
cost
enables
confidential
enable
field
attend
idea
sectors
success
learn
planning
deliver
provided
trainee
productive
behavior
performing
experts

factors typically associated with human capital accumulation do not significantly affect demand for these types of programs. The positive correlation between WTP and income aligns with our expectations, reinforcing the notion that willingness to pay increases with financial means.[16]

A key question arises: what explains the relatively low demand for professional development opportunities, especially given the widespread acknowledgment of their potential benefits? To explore this, we investigate the reasons behind the low demand, we analyzed respondents' stated intentions behind their participation in professional development. We find that private motivations—such as career advancement—are negatively correlated with WTP. In contrast, public or collective motivations—such as contributing to social impact or benefiting the team—are positively correlated with WTP (Figure 4). The details of the analysis are in Appendix Annex C, but in summary we use an individual's highest offer across the three capacity building initiatives as the dependant variable. We create indicator variables that capture perceptions of reasons for pursuing professional development elicited by respondents after the BDM experiment, where respondents could select more than one reason. This pattern suggests that the individuals who view the training as a means to contribute to a broader cause are more willing to pay for it than those who see it as a path to personal gain.

3.1 Planned follow-up experiment

To deepen our understanding of demand elasticity, we plan to conduct a follow-up survey with the same sample. In this phase, respondents will be reminded of their initial valuations and informed about an upcoming round of targeted training. We will vary the emphasis placed on the type of skills offered in random order, asking respondents to reassess their WTP based on four different framings:

- General skills for careers in the (public or private) sector
- Skills that enhance social impact
- Skills that benefit the respondent's team or organization

[16] Appendix Annex D also shows that while individuals reported WTP across activities is positive and above 0.6, there is still variation within individual, such that respondents did not just report the same valuation for each activity.

Additionally, respondents were exposed to an information recording that emphasized different scopes of the training. By comparing WTP across these different framings, we aimed to assess whether varying the perceived benefits—either private or public—would influence demand.

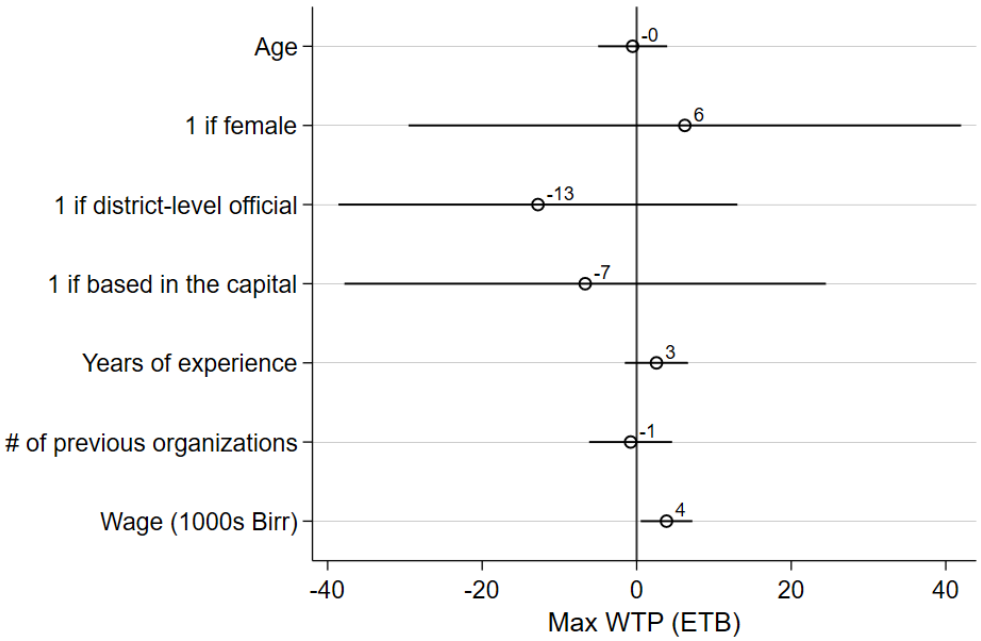


Figure 3: Relationship between individual characteristics and willingness-to-pay

4 Individual Incentives for Building the Capacity of the State

To embed the results of our field experiment into a broader view of public sector capacity building, we draw on surveys of public servants from across a range of other countries. The Global Survey of Public Servants brings together large-scale, harmonized, and comparable survey data on public servants from around the world [Schuster et al., 2023]. The survey measures employee attitudes and experience of management practices applicable to

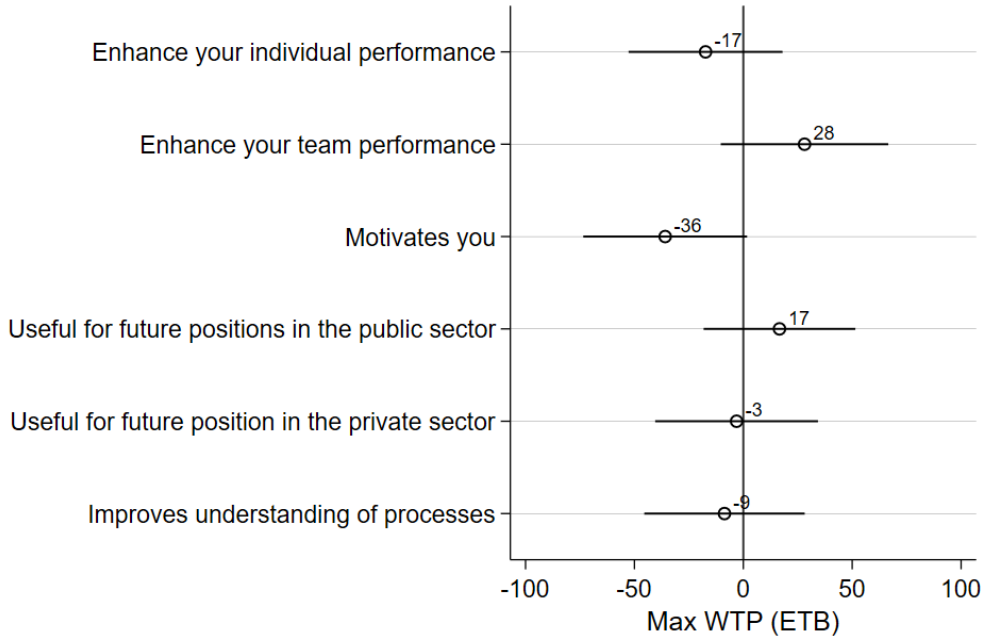


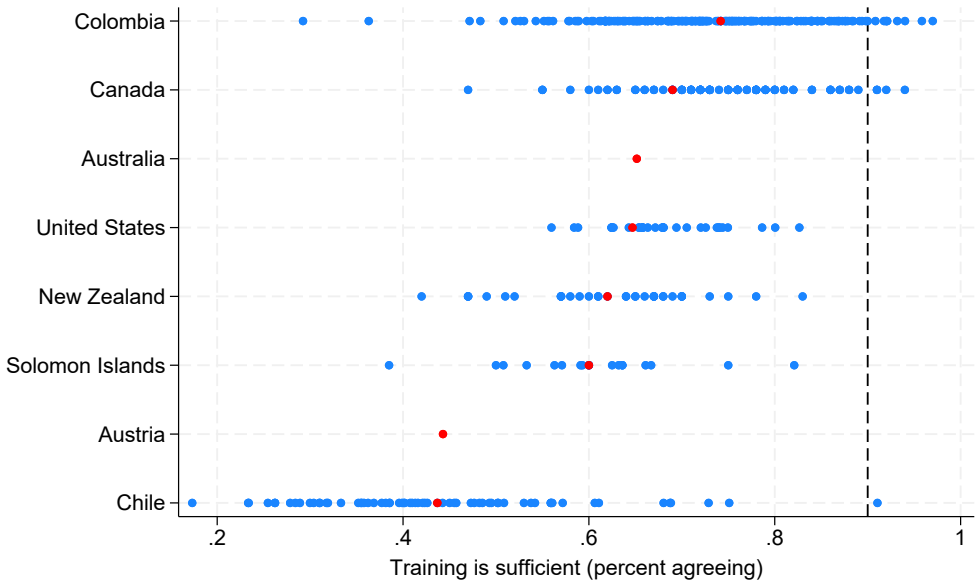
Figure 4: Relationship between reasons for capacity building and willingness-to-pay

capacity building.[17] As such, it provides a means of assessing whether the dynamics identified in the Ethiopian setting are prevalent in public administrations elsewhere.

While we do not have willingness-to-pay estimates across countries, we find many of the institutional features elaborated above in the comparative data. Training is a pillar of public service life. On average, over 54 percent of public servants reported attending a training program in the previous two years. Yet demand is high at low prices, consistent with our results. A large proportion of public servants globally do not feel there are adequate subsidized training opportunities. Figure 5 shows the distribution from the countries where the surveys asked about whether subsidized training provision is sufficient. On average, only 60 percent of officials interviewed agree that training offerings at low prices are sufficient, while just 17 percent of surveyed institutions across

[17]Further details on the data used in this section is available at the Global Survey of Public Servants web site: www.globalsurveyofpublicservants.org.

the sample have over 80 percent agreeing. Large groups in all countries and in all institutions within countries think the scale of subsidized training offered is inadequate.



Note: Sourced from the Global Survey of Public Servants. Average percent of respondents agreeing that training is sufficient are shown at the national level (red) and organization level (blue) by country.

Figure 5: Perception of adequate training (Institution level)

Indeed, the evidence shows that professional development is unequally distributed. We see substantial variation by organization in training rates. Romania, for example, has organizations with 1 percent of respondents attending training, and other organizations with 89 percent attending training. Lithuania, with the highest national average, still shows institutions varying between 45 percent and 94 percent. Such stark variations pose an important question regarding the efficiency of existing professional development activities.

The global survey evidence suggests civil servants are effectively recognising training needs, highlighting that the uneven allocation of professional development may be hampering state capabilities. The global survey data show a positive correlation in the level of public servant perceptions of having the

skills they need for their job, and perceptions that training is sufficient. In four countries with comparable data, there is a strong ($\rho = 0.489$) and highly significant ($p < 0.000$) correlation between perceptions of skills and adequate training (where we account for country-level fixed effects): civil servants in institutions with low perceived skill adequacy are also saying training is insufficient, suggesting that adequate training is a foundation for a high skilled public service.

5 Discussion

This paper investigates the demand for public service capacity building activities amongst public servants. We implement a Becker-DeGroot-Marshak mechanism adapted from the recent literature to our setting. We find low latent demand for such activities, with 'training' slightly more valued than coaching or work shadowing. The demand curve is, however, highly elastic, and at low levels of individual disutility—a low cost of access and limited personal financial burden—the vast majority of public servants will take up capacity building opportunities. Our study contributes to the growing literature on demand for continuous professional development, complementing existing evidence from the private sector [Maffioli, McKenzie and Ubfal, 2022; Abebe et al., 2024].

Descriptive evidence indicates that the individual demand is higher when individuals report public or collective benefits. Future planned research aims to disentangle whether the low demand is driven by low private returns to training or by externalities that respondents fail to internalize (see Appendix Section E). Understanding this distinction is crucial for designing policies that can more effectively stimulate demand for skill-building in the public sector.

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Online Appendix

Annex A: Survey Data

Survey evidence is drawn from three sources: i) the Ethiopian Survey of Public Servants 2016, ii) the Ethiopian Survey of Public Servants 2024, and iii) the Global Survey of Public Servants dataset.

Global Survey of Public Servants. The Surveys of Public Servants aim to gain microdata on the perceptions and experiences of civil servants, on the key obstacles civil servants face as they perform their duties to the best of their abilities, and ultimately on the provision of public services.

The global data was used in Section 4, and provides large-scale, harmonised, and comparable data sets across many countries on the perspectives of civil servants at the institutional level. More information about the question modules, methodology, and coverage is availalble at the GSPS website: www.globalsurveyofpublicservants.org.

Two such surveys been conducted in Ethiopia, where we have individual-level data. The interviews were carried out in the strictest confidence to ensure accurate responses. Enumerators signed and presented confidentiality agreements at the start of the interview, which all began with a full explanation of the research protocols established to ensure respondent anonymity.

Ethiopia Survey of Public Servants 2016. This data was used to form the baseline representative group to compare our willingness to pay sample to in Section 2. A survey of $N = 2,164$ face to face interviews was implemented in 2016. The survey utilized random sampling techniques at all levels of sample selection to gain a representative sample of officials across the government sectors studied. Interviews were carried out in June to September 2016 across the three tiers of Ethiopian government; at the federal level, in all 11 regions existing at that time, and in a selection of 66 woredas. Civil servants in non-managerial positions (employee-level track), civil servants in managerial positions (director-level track), and organization heads (political appointee-level track) were included. Frontline service delivery staff such as teachers or nurses were not included.

Ethiopia Survey of Public Servants 2024. This data was used for information on current Ethiopian civil servants perspectives on and experience

of professional development in Section 2.1. A survey of $N = 3,321$ face to face interviews was implemented in February to April 2024 across the three tiers of Ethiopian government. Civil servants in non-managerial positions (employee-level track), civil servants in managerial positions (director-level track), and organization heads (political appointee-level track) were included. Frontline service delivery staff such as teachers or nurses were not included. The survey especially sampled civil servants in the sectors of Finance, Revenue, Labor and skill, Health, Education, Agriculture, and the Civil Service itself.

Annex B: Willingness-To-Pay Elicitation Procedure

Non-payment

Non-payment interferes with establishing the willingness to pay in a real situation, and if participants anticipate they need not actually pay, may not reveal their true demand. We have followed best practice in implementing the willingness-to-pay mechanism in an attempt to reduce the risks of non-payment. Existing studies have wide variation in the extent of non-payment. Maffioli, McKenzie and Ubfal [2022] experienced a high degree of non-payment, with over half refusing to pay the full amount assigned to them for the training. Studies that have requested payment at the same time, or on the same day, as eliciting willingness to pay experienced much less default [Berry, Fischer and Guiteras, 2020; Burchardi et al., 2021]. Maffioli, McKenzie and Ubfal [2022] review non-payment and find that most studies report single-digit percentages of reneging. By requesting payment upfront, we mitigated the amount of non-payment. Messaging the need to pay may itself reduce the possible expectation that non-payment is an option.

Sale Mechanism Description

Enumerators read out the following.

1. We would like to offer you professional development opportunities. We want to get a sense of its value to you and others, so we are not going to tell you the price yet.

2. We are going to do an activity to get a sense of your interest and then we'll tell you the cost. To make it concrete, you will only be able to

buy the professional development opportunities if you've already given a high enough price. We will explain this in more detail.

3. The price you pay will be determined by this tablet. You will pay the amount on the tablet and **the amount you tell us you are willing to pay will only determine *if* you can buy these things, not the amount you pay.**
4. We understand this is an unusual way to sell an item or a service, but recording your answer in this way helps us understand how much people value the service.
5. We will offer [Coaching, Shadowing and Training][18] as professional development opportunities. We will ask you about all three at the same time.
6. You will not have to spend any more than you really want to.
7. You may even be able to buy them for less.
8. You will only buy one opportunity at most.
9. Here is how the promotion works:
   (a) I will ask you to tell me the maximum price you are willing to pay for each of [Coaching, Shadowing and Training].
   (b) You will decide how much you will be willing to pay for them at the same time.
   (c) The tablet will randomly draw one of the three opportunities, and then the tablet will draw a random number.
   (d) This number will be the price for the development opportunity.
   (e) If your bid is lower than the numbers shown on the tablet, then you cannot buy the development opportunities.
   (f) If your bid is greater than or equal to the number the tablet generates, you will purchase the professional development opportunity and you will pay the price shown on the tablet.
   (g) You will attend only one opportunity at most.

[18] Indicates randomised order

(h) You will only have one chance to play for the opportunities.

(i) You cannot change your bid after you draw from the tablet.

(j) You must state a price that you are actually able to pay today.

10. Can you access TeleBirr?

    (a) *If no:* Do you have cash available now or are you able to get it today?

    (b) *If no:* If you are not able to access telebirr or have access to cash today, please contact a facilitator.

11. We will practice in one moment. At any moment, you may contact a facilitator if you have questions or need assistance.

Practice Sale

2. a. *Show pen.*

   b. What is the maximum you are willing to pay for this?

      i. *They say an amount X.*

   c. If we draw a number that is smaller than (or equal to) [X], you will pay the amount drawn by the computer and receive the pen.

   d. If we draw a number that is greater than [X], you will not be able to purchase the item, even if you later decide you would be willing to pay the greater number. You cannot change your bid after you pick a price. Do you understand?

   e. Please tell me, if you bid [X] as the amount that you are willing to pay and we draw [X+20 Birr], what happens?

      i. *Repeat explanation if incorrect.*

   f. And if you bid [X] as the amount that you are willing to pay and we draw [X-20 Birr], what happens?

      i. *Repeat explanation if incorrect.*

   g. If we actually drew [X+20 Birr], would you still want to purchase the item?

i. *If yes, repeat this step.*

ii. *If no, move on.*

h. So, is [X] truly the most you would want to pay?

i. If you pick [X], you must be able to pay [X]. Are you able to pay [X] today?

i. *If no, ask for maximum they are willing and able to pay now.*

j. *Record maximum amount [X].*

k. Do you have access to TeleBirr, and are you able to pay the amount you have stated you are willing to pay using TeleBirr?

i. *If they do not have access to TeleBirr, ask:*

Do you have enough cash to pay the amount you have stated you are willing to pay? Could you show it to me?

ii. *If they cannot present the cash, ask:*

Will you be able to fetch the cash by the end of today?

iii. *If they cannot, confirm that they must say the highest amount they are willing and able to pay.*

l. Please enter the maximum amount in the response form.

m. Now you will draw a price using the computer's random number generator. If you have picked a price larger than the one drawn, you will buy the item at the price the computer draws. If you have picked a price smaller than the amount drawn, you will not be able to buy the item. Are you ready to draw the number?

n. *Respondent clicks button and price is recorded.*

o. *Whether price drawn is higher than [X] (no sale) or price drawn is less than or equal to [X] (sale) is displayed. If sale, exchange item for cash.*

p. Do you have any questions about the game?

Executive Leadership Institutes

Show advert for professional development opportunities. The advert will be based on the following information:

1. **Opportunities for Executive Education from International Partners**
   Would you like to accelerate your executive learning? In collaboration with university leadership, staff from the Universities of Oxford and Cornell and the World Bank are collaborating to provide a set of executive education opportunities to Civil Service University students over the next few weeks.

   We will now describe these opportunities to you.

   - **Training:** Learn about some of the biggest new ideas in working in public service from specialists in the field. Over an intensive one-day course, you will be taught key insights from research on the Ethiopian Government and hear about best-practice from some of the most successful managers in Federal Government.
     Logistics:
     - [DATE]
     - Full day training course at CSU
     - Topics: Overcoming the biggest obstacles in Ethiopia's public service; building your ability to contribute to Ethiopia's national priorities; becoming a better executive and manager
   - **Coaching:** Would you like a personal assessment of your approach to work with a group of public sector specialists and successful managers? You will be supported to outline key components of your approach to work and then reflect on these with a group of trained experts in the field. This activity gives you tailored feedback on how you might strengthen your professional performance.

     Logistics:
     - [DATE]

- Sign up for any 3-hour session split between reflection, group talk, and individual action planning
- Opportunity for follow-up check-ins with effective members of the service

- **Work shadowing:** Shadow an effective member of service staff to see exactly what it takes to be successful in the Ethiopian public service. You will have the opportunity to observe real-world examples of how to operate successfully within the service and discuss your observations with a small group of other trainees. This activity gives you practical examples of how to strengthen your professional performance within the Ethiopian public service.

  Logistics:

  - Elect an afternoon in the week of X that you would be available to undertake the work shadowing
  - Small groups of 5 trainees shadowing a successful officer at the Ethiopian Management Institute
  - A combination of observation and small-group discussion of key strengths observed

Survey Instructions - BDM Mechanism

1. a. What is the maximum you are willing to pay for the [course]? Please also write the amount down on some paper.

   i. *If positive:*

      A. If we draw a number that is smaller than (or equal to) [course offer], you will pay the amount drawn by the computer and participate in this professional development opportunity.

      B. If we draw a number that is greater than [course offer], you will not be able to pay (or receive money) to participate in this professional development opportunity, even if you later decide you would be willing to pay the greater

number. You cannot change your bid after you pick a price.

C. Do you understand? Please contact a facilitator if you have any questions.

ii. *If negative:* Perhaps you would be willing to attend the opportunity if you received money, instead of paying for it.

What is the minimum amount of money you would need to receive to attend the [course]?"

A. You have indicated we would have to pay you to attend the professional development opportunity.

B. If we draw a number that means we pay you more than (or an amount equal to) [course offer], you receive the amount drawn by the tablet and participate in this professional development opportunity.

C. If we draw a number that means we would have paid you less than [course offer], or that you would need to pay something for it, you will not be able to participate. This is true even if you later decide you would be willing to receive less or pay something. You cannot change your bid after you pick a price.

D. Do you understand? Please contact a facilitator if you have any questions.

b. If we actually drew [course offer + 20] Birr as the amount you pay, would you still want to purchase the opportunity?

c. *If yes, repeat steps*

d. If you pick [course offer], you must be able to pay [course offer]. Is [course offer] truly the most you would want to pay, and are you able to pay it today?

e. *If no:* What is the maximum you are willing and able to pay now?

Repeat in random order for the three course types. Then show all three bids on one page on tablet and ask:

2. Looking at all three opportunities, are you happy with the amounts you bid? Remember, you will only need to pay for at most one of them.
3. The maximum you have stating you are willing to pay for any opportunity is [max bid].
4. Do you have access to TeleBirr, and are you able to pay the amount you have stated you are willing to pay using TeleBirr?
   (a) *If no TeleBirr:* Do you have enough cash to pay the amount you have stated you are willing to pay? Could you show it to me?
   (b) *If insufficient cash on them:* Will you be able to fetch the right amount of cash by the end of today?
   (c) *If unable to fetch sufficient cash by the end of the day:* Please return to the bid summary and reduce your bids. You may also talk to a facilitator.
5. Now you will draw a price using the tablet's random number generator. First, we will randomly draw one of the three opportunities.

   If the price you are willing to pay is below the one randomly drawn by the tablet for this opportunity, you will buy the opportunity at the price the computer draws.

   If you have picked a price smaller than the amount drawn, you will not be able to buy the opportunity.

   Are you ready to draw the number? The next question will draw the randomised price.

Remaining Survey Questions

1. Respondent completes remaining questions on tablets.

Annex C: Regressions on Willingness to Pay

For the analysis presented, we used the maximum bid for any training type, winzorized at the 99th percentile, as the dependant variable. The variable is displayed in the Figure B1 alongside the individual valuations for each capacity-building activity.

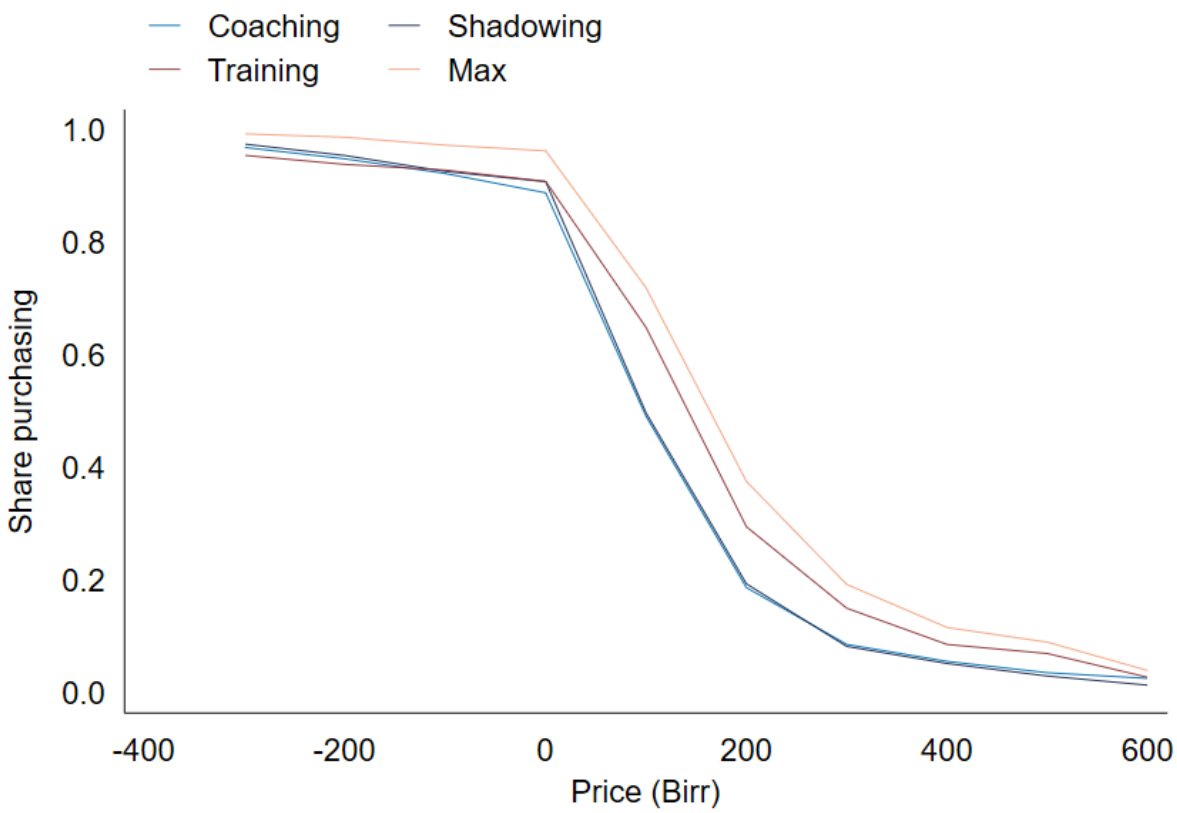

Figure B1: Inverse Demand Curve for Coaching, Shadowing and Training; Including the Maximum Bid

For each variable, we coded a binary indicator, assigning a value of 1 if the respondent's reported that their professional development was relevant for the following (non-mutually exclusive) reasons, and 0 otherwise:

- Enhancing their individual performance
- Enhancing their team's performance
- Motivating them
- Being useful for future positions in the public sector

- Being useful for future positions in the private sector
- Improving their understanding of processes

We conditioned on the demographics variables we collected data on: age, whether the respondent had tertiary education, the time they had worked in the public service, whether they worked in a decentralized (woreda) tier of government, and a log of their reported wage. Standard errors are robust.

Annex D: Bid Correlations

Spearman's rank correlations between bids.

	Coaching	Training	Shadowing
Coaching	1.00		
Training	0.70	1.00	
Shadowing	0.65	0.68	1.00

Note: Non-winsorised data with casewise deletion, $N = 496$. All correlations significant with $p < 0.000$.

Annex E: Pre-Analysis Plan

At the time of writing we are completing a follow-up survey by phone with an embedded information provision experiment. Many of the students are expected to have returned to their offices, making in person elicitation less feasible. We are measuring their willingness-to-pay for the four different training emphases, treat respondents with our audio intervention, and then re-measure their WtP. The rest of this Appendix Section provides details the structure of this follow-up sruvey and our planned future analysis.

E0.1 Demand for professional development

We will present four kinds of training emphases to our professional student sample, enabling us to discriminate between demand for each. The prompts will be read in case the subject asks for further information.

> **Scenario:**
> We are thinking of offering another one- or two-day training that

emphasizes [X]. How much would you actually pay for that kind of training?

PROMPT: What's the least amount you'd have to be paid to take such a course?

[**Randomize order of presentation**]

1. **General skills for careers.** ...skills to strengthen your career prospects across any kind of organization.
   - *PROMPT: The kind of training that enhances your career by learning skills you can take to any organization and increase your performance level there. You will learn tools and strategies to be more effective in managing people and projects.*

2. **Spillovers to the organization.** ...skills that are identified by your team as crucial for your organization's productivity.
   - *PROMPT: Do you want to learn and practice how to make your team work together more effectively, reduce bottlenecks, and increase the part you can play in successfully delivering your team's objectives?*
   - *PROMPT: Emphasizing complementarities in how people work. You can reduce bottlenecks and improve how your team works together and with other teams.*

3. **Spillovers to service as a whole.** ...skills that make the public service as a whole work more effectively together.
   - *PROMPT: The public sector has specific requirements to succeed. Learn about public sector leadership, navigating civil service rules and culture to progress your career.*

4. **Delivering social impact.** ...skills that make your direct impact on citizens larger.

- *PROMPT: This training will aim to provide new ways of working to help you directly deliver better outcomes for citizens.*

For the follow-up phone survey, we will mimic the BDM mechanism employed in the first part of our study in November 2023, while stating that we will not collect money on that day. We state that if we are to go ahead with the proposed training, the amounts they state will be the amounts they may have to pay. This allows us to measure WtP in a way that is consistent our first experiment in November 2023, but investigate many more variations in the characteristics of training in the finite interview time.

E0.2 Survey experiments

To investigate the determinants of WtP for public service capacity building, we ask participants to assess their WtP in the case that training emphasises each of the themes outlined above. We then undertake two randomised survey experiments:

Testing saturation: We measure the extent to which individuals change their WtP conditional on the fact that other members of their unit are being trained. Specifically, we randomise the number of individuals that we state will be trained with the participant. The question line is as follows:

> Thank you. We are interested in whether the fact that from your department, [RANDOMIZED: 1 other; 2 others; 3 others; the rest of your department] going to a training affects your willingness to participate.
>
> Imagine that from your unit, [RANDOMIZED: 1 other; 2 others; 3 others; the rest of your department] registered to attend the training that emphasizes [RANDOMLY CHOOSE ONE: skills to strengthen your career prospects across any kind of organization; skills that are identified by your team as crucial for your organization's productivity; skills that make the public service as a whole work more effectively together; skills that make your direct impact on citizens larger].
>
> Without that consideration, you had bid [X] Birr. How much are you willing to pay now that [RANDOMIZED: 1 other; 2 others;

3 others; the rest of your department] are attending?

Testing emphasis: We prime the participant towards a private career intention for training, or public contribution intention, by having them listen to a recording from the Civil Service Commission that describes why training is important. The text of these recordings are provided in Table 1. Which of the two recordings is played to the participant is randomised. The audio recording is introduced by the enumerator as from the Civil Service Commission, with whom we have collaborated. These are read by a female ex-senior civil servant to contrast with the male enumerator, differentiating the message from the call.

Table B1: Audio Treatments

Treatment	Description
Private—Career	"The Civil Service Commission wants to support the careers of civil servants, and so are implementing a competency framework to make clear the steps needed for career advancement. Training is a central part of our reform process to help all civil servants reach these standards. We want civil servants to perform and progress in their own personal careers."
Public—Other-regarding	"I've seen first-hand how training can make a difference in the work civil servants do. I recently visited a team in the service, and saw how by implementing the new practices they learnt at a training course, they made their processes much more efficient, and were able to serve more members of the public. Citizens I spoke to were very happy about their experience of the team, which is the feedback I love to hear. Improving skills and ways of working through training can make a difference for Ethiopia."

E0.3 Assignment to Treatment

Assignment to treatment in the survey experiments will be based on a pre-assigned randomization. Participants will be linked to this randomization with their survey ID. The phone call order is further randomized.

E0.4 Attrition from the Sample

To ensure as small an attrition rate as possible, we will use the following procedure. Firstly, participants are called and invited to conduct the 15-minute survey at that point, or else reschedule for a more convenient time. Secondly, in case they cannot be reached, the enumerator will call them up to three times. Thirdly, they will be offered alternative methods, such as zoom calls, to conduct the survey. We provided respondents with a 50ETB (0.50USD) in airtime credit to thank them for their time.

E1 Empirical Analysis

E1.1 Variables

E1.1.1 Independent Variables

Testing saturation

C is a vector of indicators variables equal to one for each of the respective options "1 other"; "2 others"; "3 others"; "the rest of your department". The omitted vector in the regression will be "1 other" colleague.

Testing motivation

T_i is an indicator variable corresponding to audio message, defined as: 0 = career incentives message, 1 = citizen impact message.

E1.1.2 Outcome Variables

Primary outcome: The key outcomes of interest are the elicited demands for professional development, asked for each of the four types of training, or the randomly selected one after the survey experiment on saturation.

Secondary outcome: We will ask participants about their anticipated promotion prospects with the following question:

> Thank you. We're interested in how training impacts your chances of promotion. Imagine you went on a training that emphasizes [RANDOMLY CHOOSE ONE: skills to strengthen your

> career prospects across any kind of organization; skills that are identified by your team as crucial for your organization's productivity; skills that make the public service as a whole work more effectively together; skills that make your direct impact on citizens larger]. Do you think that increases the percentage chance of receiving a promotion? By how much? [ANSWER OPTIONS: 0% - 100%]

We also collected data on whether the respondent has been promoted in the last year that we plan to use as an additional control variable ("Are you promoted since attending CSU? 1 = Yes, 0 = No").

E1.2 Specifications

E1.2.1 Demand Curves

We will estimate and plot demand curves for the four training types and compare them with demand for training from the initial elicitation in November 2023.

Nonparametric tests of equality of distribution will be run to assess whether there are differences between the estimated demand curves for each type of training.

We will also estimate the mean WtP for each of the four trainings in our initial elicitation. To test for differences, we will regress WtP in the emphasis $\mathbf{S}$. Standard errors will be robust for each individual i.

$$Y_i = \mathbf{S}\boldsymbol{\beta} + \varepsilon_i \tag{1}$$

E1.2.2 Survey experiments

Testing saturation.

We will run the following regression to test the impact of the number of colleagues attending training, where we control for the (randomly selected) training emphasis S.

$$Y_{post,i} = \alpha_1 Y_{pre,i} + \mathbf{C}\boldsymbol{\gamma} + \mathbf{S}\boldsymbol{\delta_1} + \varepsilon_i \tag{2}$$

where $Y_{pre,i}$ is the elicited willingness-to-pay elicited at the beginning of the survey, for the corresponding emphasis.

Testing motivation.

We will run the following regression to test the impact of the audio-message, where we control the (randomly selected) training emphasis S.

$$Y_{post,i} = \alpha_2 Y_{pre,i} + \beta T_i + \mathbf{S}\boldsymbol{\delta_2} + \varepsilon_i \tag{3}$$

where $Y_{pre,i}$ is the elicited willingness-to-pay elicited at the beginning of the survey, for the corresponding emphasis.

We also plan to run an alternative difference-in-differences specification if we find an autocorrelation above 0.8 between the pre-treatment and post-treatment elicited willingness to pay.

Secondary outcome: To study changes in the perceived probability of promotion from training, we will use a specification similar to Equation 3, but controlling for an indicator for the individual has been previously been promoted, rather than the prior value $Y_{pre,i}$.

E1.2.3 Controls

We will estimate equations (1), (2), (3) under the following conditions:

1. Controlling for the willingness-to-pay for training elicited in November 2023.
2. Using a post-double LASSO model for control variable selection.

The set of controls included in the post-double LASSO selection will include: age, sex of the respondent, indicators for the region in which the respondent, indicators for which tier of the administration, the years of experience in the civil service, number of organizations that the respondent has worked in, indicators for monthly income brackets before starting the course at CSU.